General comments on EE4-25 Design of Linear Multivariable Control Systems 2019

Overall, the students have done well on the exam paper, scoring an exam average of 77%.

1. The students have done less well on this question, scoring approximately 58%. This is a robust control design question with a rather complex uncertainty description and is a more elaborate variation on past exam questions.
   (a) Tests the ability to correctly state the small gain theorem.
   (b) Tests the ability to translate the uncertainty description into design specifications.
   (c) Tests the ability to translate the robustness specifications into performance specifications.
   (d) Tests the ability to embedd the performance specifications into the generalised regulator problem.
   (e) Tests the ability to analyse the performance properties of the robust design.

2. The students have done very well on this question, scoring approximately 86%. This question concerns the statement, derivation and application of the bounded real lemma.
   (a) Tests the understanding of the statement and derivation of the bounded real lemma and a variation that is linear in the system matrices through the use of Schur complements.
   (b) Tests the application of the bounded real lemma in a disturbance rejection design using output injection.

3. The students have done well on this question, scoring approximately 71%. This is linear quadratic regulator design problem posed in a form which is close to the generalised regulator problem. It is a variation on past exam questions.
   (a) Tests the ability to derive the solution (controller and minimum cost) in terms of the solution of a Riccati equation.
   (b) Tests the ability to prove closed loop stability and the assumptions needed to guarantee this.
   (c) Tests the ability to analyse the properties of the solution, in this case the minimum phase property.

4. The students did really well on this question, scoring approximately 92%. This is a generalised regulator design question for a simple state–feedback problem.
   (a) Tests the ability to formulate the generalised regulator problem for the state–feedback problem.
   (b) Tests the ability to reformulate the problem into a tractable problem in terms of a Lyapunov function and using completion of squares.
   (c) Tests the ability to reduce the reformulated problem into the solution of a Riccati equation.
   (d) Tests the students' understanding that the linear quadratic regulator problem in Question 3 is a special case of the generalised regulator problem.